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REVIEW

Quantum sensors for enhanced positioning and navigation: a comprehensive review

Haobo Li[1] · Safoora Zaminpardaz[1] · Allison Kealy[2] · Andrew D. Greentree[1] · Eldar Rubinov[3] · Brant Gibson[1] · Suelynn Choy[1]

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Abstract

The inherent vulnerability of Global Navigation Satellite Systems (GNSS) to interference, spoofing, and signal degradation highlights the urgent need for robust augmentation and alternative solutions to support resilient Positioning, Navigation, and Timing (PNT) services. Among emerging technologies, quantum sensing stands out for its ability to harness fundamental quantum phenomena, such as superposition and entanglement, enabling absolute, drift-free and ultra-sensitive measurements of inertial forces, gravity, and magnetic fields with long-term stability and reduced dependency on external signals. This capability offers a promising pathway to enhance positioning and navigation performance in GNSS-denied or GNSS-contested environments. Here, we present a comprehensive review of quantum sensing technologies in the context of positioning and navigation, including cold atom interferometry, atomic vapour cell, superconducting quantum interference devices, and room temperature solid-state systems like nitrogen-vacancy centres in diamond, and their potential to support positioning and navigation across a range of operational contexts. Both standalone and hybrid quantum-classical architectures are examined, with case studies illustrating their ability to reduce navigation drift and achieve sub-metre accuracy through quantum-enhanced inertial navigation, gravity-aided positioning, and magnetic field-matching approach. Despite their promise, challenges including decoherence, environmental robustness, miniaturisation, and integration complexity remain significant barriers to practical deployment. Rather than replacing existing classical systems, quantum sensors are best served as complementary solutions within multi-layered PNT frameworks. This review also identifies priorities for advancing field-deployable quantum positioning and navigation solutions, including sensor robustness, miniaturisation, seamless integration, and the development of protocols to quantify quantum performance. Aligned with international quantum initiatives, this work contributes to the vision of secure, interference-resistant navigation infrastructure for scientific, industrial, and defence applications.

Keywords Resilient positioning and navigation · Alternative PNT technology · Quantum sensing · GNSS-denied environment.

Introduction

Positioning, Navigation, and Timing (PNT) capabilities have become deeply integrated into the fabric of modern society, underpinning critical operations across many sectors (McNeff 2010; Detratti and Dolce 2020). As these technologies evolve, new applications continue to emerge for positioning and navigation, particularly in consumer technologies and autonomous systems (Smith and Johnson 2013; Morton et al. 2021). In addition, accurate timing remains especially crucial as the backbone for essential infrastructures such as financial systems, communication

✉ Haobo Li
haobo.li@rmit.edu.au

[1] School of Science, RMIT University, Melbourne, VIC 3001, Australia

[2] Innovative Planet Research Institute, Swinburne University of Technology, Melbourne, VIC 3122, Australia

[3] FrontierSI, Melbourne, VIC 3000, Australia

networks and power distribution grids (Jones 2017; Piester et al. 2019).

Global Navigation Satellite Systems (GNSS), like the Global Positioning System (GPS) and Galileo, are the primary PNT technologies driving the expansion of PNT applications (Yang et al. 2011; Li et al. 2015). However, the central role of GNSS in modern PNT systems has introduced a critical vulnerability: it constitutes a single point of failure within the PNT infrastructure. As a result, any disruption caused by intentional or unintentional interference, and partial or complete signal outages, can propagate widely, affecting all dependent domains (Ioannides et al. 2016; Elghamrawy et al. 2020). To address these vulnerabilities, a range of technologies, sensors and techniques, either as augmentations to GNSS or as alternative solutions, have been developed to enhance the overall robustness and adaptability of PNT services (Kealy et al. 2015; Retscher et al. 2020).

We are experiencing the second quantum revolution (Dowling and Milburn 2003). This transformative phase marks the shift of quantum systems from fundamental physics experiments confined to laboratories to practical implementations in real-world technologies. In this rapidly evolving landscape, exploring the impact of quantum systems on PNT is timely and important (Kealy and Greentree 2025). While quantum mechanics accurately describes the fundamental laws governing particles and their interactions, the term "*quantum*" is not always consistently or clearly defined in the context of sensing. Technically, all sensors operate within the quantum realm, however, classical sensors are considered "*trivially*" quantum because their operation can be adequately explained using classical physics. This conceptual distinction therefore forms the basis for differentiating classical from quantum sensors. For example, a fluxgate magnetometer is a classical device that relies on the response of magnetic cores, governed by Maxwell's equations and characterised by continuously varying parameters. In contrast, quantum sensors, such as superconducting quantum interference devices (SQUIDs) (Fagaly 2006) or atomic vapour cell magnetometers (Ma et al. 2024), exploit properties of quantum physics that cannot be completely represented by classical theories. In general, quantum sensors can be defined as devices that exploit one or a combination of the following characteristics: (1) nanoscale dimension; (2) quantum protocol; (3) quantum advantage; (4) reliance on fundamental constants rather than calibration. Specifically, a single fluorescent nanodiamond exemplifies a nanoscale quantum sensor. A quantum protocol refers to a structured sequence of quantum operations designed for enhanced sensing performance, such as Kitaev's phase estimation algorithm applied to measurements of unknown optical phases (Kitaev 1995; Higgins et al. 2007). Quantum advantage describes the superior scaling achieved using non-classical states, like superposition states in cold atom interferometers or optimally entangled states as described in (Berry and Wiseman 2000), which are not accessible to classical states. Finally, reliance on fundamental constants rather than calibration characterises systems that leverage the identical properties of atoms of the same type, as demonstrated in ion clock systems (Ludlow et al. 2015).

Quantum sensors offer several critical advantages, including novel sensing modalities, compact and ultimately low-power operation, and unparalleled sensitivity constrained only by the fundamental laws of physics (Degen et al. 2017). These attributes collectively position quantum technologies as potential game-changers for next-generation PNT systems. However, despite their promise, quantum technologies remain in a development phase compared to well-established classical solutions. The extreme sensitivity of quantum sensors, while advantageous, also makes them highly susceptible to extraneous noises that can masquerade as signals of interest (Budker and Romalis 2007). Moreover, entanglement, as the most powerful quantum resource, characterised by non-local correlations between particles not explained by classical physics, is inherently fragile and challenging to maintain (Dür and Briegel 2004). This fragility stems from both the sensitivity of quantum systems to environmental influences such as electromagnetic fields and thermal noise, and the fact that measuring any part of an entangled state collapses the quantum mechanical wavefunction, thereby disrupting all particles in the entangled system (Yu and Eberly 2002; Sanders 2012). Another challenge lies in the widespread use of interferometry in quantum system designs, which are susceptible to phase-wrapping issues, where the measured signals are only determined modulo 2π (Gustavson 2000a; Cronin et al. 2009). Additionally, the integration of quantum sensors with classical platforms remains complex due to mismatched operational principles (such as bandwidth and dynamic range), sensitivities, control requirements, and other specifications (Cassel et al. 2023; Vidal Bustamante 2025). Hence, achieving effective integration and seamless operation often requires extensive re-engineering and adaptation.

This paper presents a comprehensive review of quantum sensors aimed at addressing the challenges confronting existing PNT architectures. This review focuses specifically on positioning and navigation, intentionally excluding the timing aspect to allow for more concentrated exploration of these two closely related tasks. The narrowed scope enables a deeper investigation into quantum-enhanced positioning and navigation, facilitating a more detailed and rigorous analysis of their capabilities and limitations.

The rest of this paper is structured as follows: the following section reviews the current PNT landscape, focusing on the vulnerabilities of GNSS and outlining viable augmentation and alternatives, particularly for GNSS-denied environments. This is followed by a comprehensive discussion of various quantum sensing systems, highlighting their basic operating principles, advantages, and limitations. Subsequently, we present several specific quantum sensors that have shown strong potential in enhancing positioning and navigation capabilities. The paper then explores hybrid systems that integrate quantum and classical sensors, supported by implementations and case studies. Finally, this paper concludes with a summary of key insights, a discussion of current technical challenges, and an outlook on future research priorities for advancing quantum-enabled positioning and navigation solutions.

Overview of current PNT systems

In recent decades, GNSS, supported by multiple ground-based and space-based augmentation systems, have fundamentally revolutionised the delivery of global PNT solutions. These systems offer significant benefits, including global coverage, high accuracy, long-term stability, all-weather operability, and cost-effectiveness, enabling the development of robust infrastructures that continuously deliver PNT services across civilian and military sectors (Langley et al. 2017).

Vulnerabilities of GNSS as a foundational PNT technology

As global dependence on GNSS continues to grow, ensuring the robustness and continuity of the services has become increasingly critical. Although GNSS typically performs reliably under nominal conditions, they remain vulnerable to a broad range of threats that can severely compromise performance, especially in challenging environments or contested regions (Jafarnia-Jahromi et al. 2012; Gunther 2014). Major vulnerabilities include:

(1) *Interference (including jamming) and spoofing*: Due to the extremely low power of GNSS signals (typically around − 130 dBm) and the massive free-space path loss incurred over the long propagation distance from satellites to receivers, these signals are highly vulnerable to interference. For example, a 1 W interferer operating within GNSS frequency bands can disrupt signal reception over several kilometres (Volpe 2001). Additionally, because GNSS signals can be intercepted and even replicated, spoofing presents a serious threat to service integrity by transmitting counterfeit signals that mislead receivers and induce erroneous information (Psiaki and Humphreys 2016).

(2) *Signal blockage and multipath*: GNSS signals have limited ability to penetrate objects like buildings and are susceptible to multipath effects caused by reflections from surrounding surfaces. As a result, signal quality and reliability degrade considerably in environments with reflective or obstructive features, such as urban canyons or dense forests. In extreme scenarios, such as underground, underwater, or heavily shielded settings, GNSS services may become entirely unavailable (Hussain et al. 2021).

(3) *Satellite constellation geometry and system failures*: In high-latitude and polar regions, GNSS constellation geometry is often suboptimal due to lower satellite elevation angles and sparse satellite coverage. This results in poor satellite visibility, increased dilution of precision values, and degraded performance and service availability (Yastrebova et al. 2021). Additionally, satellite malfunctions, anomalies in spaceborne payloads, or failures in ground control segments can further compromise the continuity and reliability of GNSS operations.

Collectively, these vulnerabilities expose the operational risks associated with exclusive reliance on GNSS and highlight the urgent need for resilient PNT architectures that integrate augmentation solutions and alternative technologies to ensure continuous, secure, and reliable PNT service delivery, particularly in environments where GNSS access is degraded or denied (Grejner-Brzezinska et al. 2016).

To address these challenges, several conceptual frameworks have been proposed for next-generation PNT architectures, including "*Assured PNT*" (Parkinson 2014), "*Alternative PNT*" (Narins et al. 2012), "*Resilient PNT*" (Yang et al. 2019), and "*Secured PNT*" (Yang et al. 2023). While differing in terminology, these frameworks share a common goal: to develop an integrated architecture that intelligently integrates information from multiple systems and sensors to improve overall system resilience. By leveraging the complementary strengths of diverse sources to compensate for the limitations of any single system, such architectures aim to enhance service availability and continuity, extend operational coverage, and ensure the reliability of PNT services (Yang et al. 2023).

Traditional and emerging augmentation and alternatives to GNSS

Today, in addition to GNSS and their augmentations, a variety of traditional systems and emerging alternatives have

been developed to enhance PNT capabilities. Notable examples are outlined below:

Inertial Navigation Systems (INS) utilise inertial sensors to measure accelerations and rotation rates, integrating these through motion equations to provide continuous, self-contained navigation solutions (Jekeli 2023). While modern INS are commonly integrated with GNSS, INS technology predates satellite navigation by decades, with early and well-established applications in aerospace, maritime, and even in-car navigation systems (Grewal et al. 2007). Despite their independence from external signals, even high-end INS suffer from sensor drifts and biases, leading to accumulating errors over time (El-Sheimy and Youssef 2020). For example, a navigation-grade accelerometer with long-term stability of ~10 µg can induce horizontal errors of ~60 m over the 84.4-minute Schuler period without GNSS aiding (Titterton and Weston 2004).

Low Earth Orbit (LEO) PNT systems, although still relying on satellite signals, represent an emerging alternative to conventional GNSS that operate in Medium Earth Orbit (MEO) (Prol et al. 2022). Unlike traditional MEO-based GNSS architectures, typically positioned at altitudes ~20,000 km, LEO satellites operate at much lower altitudes (600–1200 km), enabling signal strengths up to 1000 times greater and improved signal penetration, thereby reducing susceptibility to interference (Lawrence et al. 2017). For example, signals from Iridium satellites at 780 km are ~30 dB stronger than GPS signals (Yang et al. 2024). In addition, their high speed and dynamic nature enable more accurate and frequent measurements, enhancing resilience to spoofing and blockage (Su et al. 2019). However, achieving global coverage and continuous service requires large-scale constellations, and LEO satellites face greater exposure to factors such as atmospheric drag, gravitational perturbations, and solar radiation, leading to shorter operational lifespans and necessitate more frequent satellite replenishment (Nwankwo et al. 2015; Reid et al. 2016). Nevertheless, these challenges are increasingly mitigated by declining launch costs and the growing commercial maturity of small satellite constellations.

Natural source-based systems, such as gravity and magnetic field sensing, are also increasingly being used in positioning and navigation through map-matching techniques that correlate sensor measurements with reference field maps (Tyren 1982; Li and Wang 2014; Wu et al. 2017; Li et al. 2022; Wang et al. 2023c). These systems are particularly valuable in GNSS-denied scenarios. However, their performance depends heavily on sensor sensitivity, map resolution, and the availability of geophysical information. Additionally, interpreting subtle variations in the field requires advanced data processing, complicating real-time use in dynamic settings (Khesin et al. 2013). These challenges are closely aligned with the main focus of emerging quantum sensing research, which seeks to enhance the sensitivity and applicability of natural source-based navigation, as further discussed in the following sections.

Ambient signals from 4G/LTE, 5G, WiFi, and other cellular networks provide alternative localisation cues by exploiting signal characteristics (del Peral-Rosado et al. 2017; Laoudias et al. 2018). These techniques can achieve positioning accuracy better than 100 m and support Assisted-GNSS functionality, particularly in environments with limited satellite visibility (Zandbergen 2009). However, these methods remain vulnerable to multipath effects, interference, and the heterogeneous distribution of base stations, necessitating continuous calibration and error mitigation to ensure consistency (Xhafa et al. 2021).

Additional alternatives also include terrestrial radio navigation systems (e.g., LORAN and eLORAN) (Son et al. 2020; Specht 2021), vision-based navigation (Lu et al. 2018), and celestial navigation systems, e.g., based on stellar observations and starlight refraction (Emadzadeh and Speyer 2011; Ning et al. 2013). These systems provide valuable redundancy, enhancing the robustness of PNT services against evolving threats (Groves 2015), as depicted in Fig. 1.

In addition to the aforementioned alternatives, as illustrated in Fig. 1, quantum sensing systems have emerged as a frontier for building resilient and multi-layered PNT architectures (Kaltenbaek et al. 2021; Krelina 2023; Kealy and Greentree 2025). Quantum sensors, operating independently of external signals, enable precise measurements of magnetic fields (Budker and Romalis 2007; Wolf et al. 2015; Bennett et al. 2021), gravity and gravity gradients (Ménoret et al. 2018; Bidel et al. 2018; Bai et al. 2023), inertial information (Fang and Qin 2012; Dickerson et al. 2013; Barrett et al. 2016; Yankelev et al. 2019), and time (Ludlow et al. 2015; Jaduszliwer and Camparo 2021; Burt et al. 2021; Zheng et al. 2025). By leveraging quantum properties, these devices are already surpassing the performance limits of classical sensors in specific applications. Nevertheless, most quantum sensors remain in early developmental stages and face multiple challenges as noted above (Degen et al. 2017; Hossain 2023). The following section presents a detailed review of various quantum sensing systems, highlighting their fundamental operating principles and performance characteristics.

Quantum systems for sensing

As outlined above, the proposed classification criteria for quantum sensors offer a practical framework for understanding quantum sensing systems based on their defining

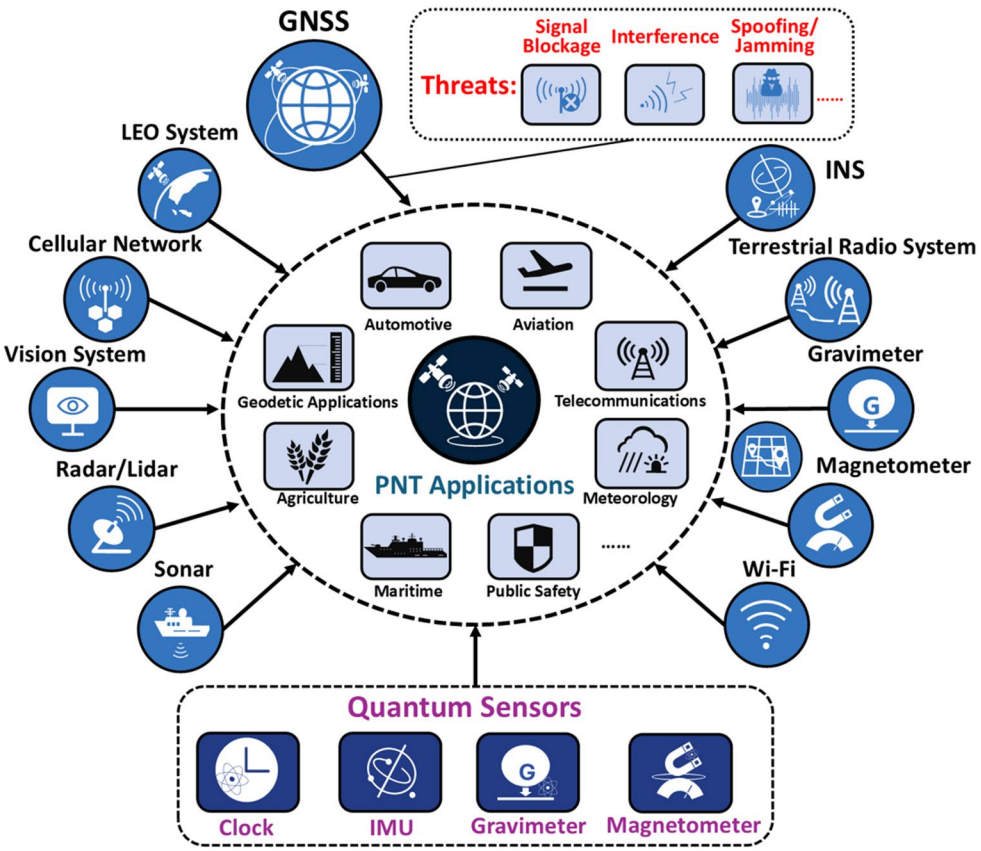


Fig. 1 Applications of PNT underpin much of modern society, including transport, agriculture, finance, meteorology, and public safety. While GNSS remain the core enabling technology, their central place in PNT infrastructure has created potential vulnerability if GNSS signals are degraded or disrupted. Numerous technologies are being explored to provide resilient PNT, with quantum sensors emerging as a promising frontier

characteristics. In the context of positioning and navigation, the criterion of nanoscale sensing is generally less applicable as a discriminator. Although measurements are made locally at the sensor, the quantities of interest, such as large-scale geophysical fields in gravitational or magnetic navigation and bulk platform rotation in quantum gyroscopes, are far beyond the nanoscale. Consequently, overall performance is governed by sensitivity at the relevant bandwidths, dynamic range, stability, and environmental robustness rather than by miniaturising the sensing element to the nanoscale. In addition, most quantum sensors currently applied in positioning and navigation rely on single-sensor units or sensor ensembles, with few implementations leveraging quantum protocols or demonstrating clear quantum advantages. Nevertheless, theoretical proposals incorporating such features are under active development and may become feasible as the technology continues to mature. At present, the key benefits of quantum sensors for positioning and navigation lie in their potential to offer absolute and drift-free measurements, reduce operational costs, and improve Size, Weight, Power, and Cost (SWaP-C) characteristics. It should be noted, however, that any SWaP-C advantage is modality dependent and not yet universal. In particular, some devices already demonstrate relatively low SWaP-C, like Nitrogen-Vacancy (NV)-centre diamond magnetometers and chip-scale atomic clocks, whereas cold atomic interferometers remain comparatively larger and more complex. Nevertheless, ongoing efforts in miniaturisation are continuously driving SWaP-C reductions. Building on these, we now examine several representative quantum systems for sensing.

The first quantum sensing system to gain widespread use was the SQUID, a highly sensitive sensor that measures electric currents circulating around superconducting loops. At low temperatures, certain materials enable electrons to pair up into composite particles known as Cooper pairs. Unlike individual fermion electrons, Cooper pairs are bosons. This bosonic nature enables them to propagate through the materials without resistance, a phenomenon known as superconductivity. The resulting current, referred to as a supercurrent, flows without energy loss and is associated with a well-defined quantum mechanical phase. The phase coherence underpins highly sensitive interference effects that are modulated by external magnetic fields (Fagaly 2006). By exploiting this property, SQUID magnetometers can detect extremely weak magnetic fields and have been widely utilised in applications such as the detection of biomagnetic signals generated by the brain (Vrba and Robinson 2002; Nishijima et al. 2013; Storm et al. 2017) and high-resolution mineral exploration (Schmidt et al. 2004; Stolz et al. 2022). Recent technological advances have further expanded their utility to PNT applications, including magnetic navigation (Djapic et al. 2015) and airborne

geomagnetic surveys (Rudd et al. 2022), which are necessary for providing magnetic maps for such navigation. Figure 2 illustrates a typical square washer SQUID architecture with an integrated input coil, exemplifying the compact and sensitive design used for high-precision sensing of magnetic fields (Schmelz and Stolz 2017; Bennett et al. 2021).

Cold atom technology, one of the most important quantum sensing platforms (Vidal Bustamante 2025), uses laser cooling techniques to reduce atomic ensembles to temperatures near absolute zero, where quantum behaviour is pronounced. Depending on the application, different temperatures are required, and hence there are different quantum solutions. For example, magneto-optical traps (MOTs) typically operate in the tens of microkelvin regime and are useful in cases requiring low atomic movement (Canuel et al. 2006; Dickerson et al. 2013), often to reduce the Doppler broadening of atomic transitions. This is the context in which they are used for atomic fountains as high precision clocks (Ludlow et al. 2015; Jaduszliwer and Camparo 2021). MOTs are also often used as the starting point for lower temperature systems. Bose-Einstein Condensates (BECs) (Hardman et al. 2016; Frye et al. 2021) typically reach nanokelvin temperatures and exploit the wave-like nature of the atomic cloud. Atomic lattices (Cladé et al. 2006; Poli et al. 2011) may either be lattices of individual particles, or small BECs in a three-dimensional optical standing wave lattice. These systems are particularly useful for accelerometry via shaken lattice or Bragg scattering (Weidner and Anderson 2018; Ledesma et al. 2025). Lastly atom lasers (Hagley et al. 1999; Bloch et al. 1999) offer outstanding coherent, travelling matter waves that are well-suited to gravimetry and accelerometry applications.

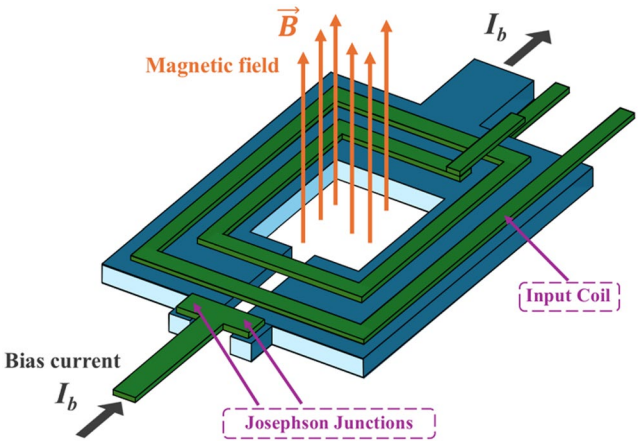


Fig. 2 Schematic diagram of a typical square washer SQUID with an integrated input coil and applied bias current. Josephson junctions are located along the edges of the slitted washer and bias current, enabling ultra-sensitive magnetic field detection through phase-coherent supercurrents. Adapted from Bennett et al. (2021), Sensors, 21, 5568, under the CC BY 4.0 license

Cold atomic interferometry (CAI) exploits quantum wave-particle duality, exhibited by BECs and atom lasers, to enable highly sensitive measurements (Cronin et al. 2009; Abend et al. 2019). In much the same way as optical interferometers sense phase shifts, CAI systems use the interference of single atoms or coherent atom waves (depending on the implementation), to sense inertial quantities like rotation and acceleration (Gustavson et al. 1997; Canuel et al. 2006; Dickerson et al. 2013; Barrett et al. 2016; Garrido Alzar 2019). In addition, CAI has been successfully employed for sensing gravitational acceleration (Peters et al. 2001; Merlet et al. 2010; Poli et al. 2011; Biedermann et al. 2015; Bidel et al. 2020) and gravity gradients (Canuel et al. 2018; Stray et al. 2022; Janvier et al. 2022).

Cold atom interferometric systems mainly include free-space (Cronin et al. 2009; Bongs et al. 2019), point-source (Dickerson et al. 2013), and guided-wave (Garrido Alzar 2019; Moan et al. 2020) interferometers. Free-space and point-source configurations, using atomic fountains or freely expanding atom clouds, prolong the interferometric time to enhance sensitivity, but they require large experimental setups. In contrast, guided-wave interferometers offer a more compact form factor, though they can be more susceptible to noise from the guiding potentials. The exceptional precision of cold atom devices stems from the inherent phase sensitivity of matter-wave interferometry (Dowling 1998; Bidel et al. 2018). This sensitivity surpasses that of optical interferometers of comparable scale due to the much shorter wavelength of the atom waves, and the markedly stronger and more direct coupling of massive particles to external fields such as gravity compared with photons. Furthermore, in light-pulse Raman atom interferometers used for gravimetry, the measured phase is independent of atomic mass, isotopic composition therefore neither introduces nor mitigates a gravimetric bias (Storey and Cohen-Tannoudji 1994; Peters et al. 2001). Nevertheless, experiments typically operate with a single isotope rather than a natural isotopic mixture, because different isotopes possess distinct hyperfine structures and Raman resonance conditions, and their simultaneous presence would lead to multiple, incoherent interferometer pathways and a loss of contrast.

In contrast to cold atom systems, atomic vapour cell technology typically operates at room temperature or elevated temperatures (Seltzer and Romalis 2009; Kitching et al. 2011). This approach utilises a gas of alkali atoms, typically rubidium or caesium, in a small cell and heats it to create an atomic vapour (Knappe et al. 2004; Camparo 2007). Although the atoms remain at or above room temperature, Doppler-free configurations, typically involving co- or counter-propagating fields, enable the selective probing of atoms with zero velocity relative to the fields (Budker and Romalis 2007). This technique ensures that

the linewidth and quantum coherence of the atomic vapour approach the limits set by cold atom systems, but without necessitating laser cooling. Spin Exchange Relaxation-Free (SERF) devices, such as SERF magnetometers, generally operate under low magnetic field conditions and at high atomic densities (Savukov 2017; Ma et al. 2024). These systems achieve exceptional sensitivity by using laser light to precisely monitor variations in atomic spin states, enabling highly accurate detection of magnetic fields (Li et al. 2018), which is closely related to magnetic anomaly map-making and magnetic anomaly-aided navigation (Bennett et al. 2021; Cassel et al. 2023).

Solid-state quantum systems, exemplified by the NV centre in diamonds (Doherty et al. 2013), involve quantum systems embedded within solid materials. An NV centre is formed when a nitrogen atom replaces a carbon atom adjacent to a vacancy in the diamond lattice, creating a point defect. Sensors based on NV centres offer several critical benefits, including compact size and excellent thermal stability, operating reliably from cryogenic temperatures up to 600–700 K (Toyli et al. 2012). NV-based sensors are notable for their utility in magnetic field measurements, as they can be optically manipulated and maintain coherent properties at room temperature. NV centres are among the leading room-temperature solid-state quantum platforms (Taylor et al. 2008; Wolf et al. 2015), but significant progress is being made in other systems like defects in silicon carbide (Koehl et al. 2011; Widmann et al. 2015) and hexagonal boron-nitride (Gottscholl et al. 2020; Chejanovsky et al. 2021), which are expanding the scope of solid-state quantum sensing technologies.

Classical optical technology for navigation is well-established, with optical gyroscopes representing one of the most mature and widely used sensors (Lefèvre 2013; Dranitsyna et al. 2023). Complementary to these classical approaches, quantum optical technology usually explores advanced phenomena such as squeezed light (Breitenbach et al. 1997; Grace et al. 2020) or entangled photons (Nagata et al. 2007; Fink et al. 2019), to perform tasks beyond the reach of classical methods (Browne et al. 2017). Quantum mechanics fundamentally sets a limit to the uncertainty with which optical parameters, such as intensity and phase, or equivalently the in-phase and in-quadrature components, can be measured. The standard quantum limit arises when measuring non-commuting parameters simultaneously, as constrained by the Heisenberg uncertainty principle. Repeated measurements in this regime typically result in precision that scales as the inverse of the square root of the number of experiments. Conversely, squeezing is the process by which the uncertainty in a given parameter is reduced below the standard quantum limit at the expense of increased uncertainty in another non-commuting parameter. Moreover, the use of entanglement (Giovannetti et al. 2004) or certain feedback mechanisms (Higgins et al. 2007) provides the best scaling possible under the laws of quantum physics. In such cases, the precision scaling follows the inverse of the number of experiments used, which is known as the Heisenberg limit. This approach increases the sensitivity of interferometers and has been successfully applied in advanced systems such as gravitational wave detectors (Goda et al. 2008). Alternatively, entangled photon states offer another pathway for enhanced sensing, enabling applications such as precise distance measurements (Giovannetti et al. 2001) and clock synchronisation (Troupe et al. 2022; Nande et al. 2025). Table 1 provides a comparative summary of the key advantages, limitations, and indicative sensitivity and modality of the available and emerging quantum sensing technologies.

As illustrated in this table, there is an increasing variety of quantum technologies, which demonstrate quantum advantage in different ways. This diversity indicates that a "one-size-fits-all" quantum approach is neither practical nor necessary. Instead, the goal for the practical use of quantum sensing systems is to identify the most suitable technology for a specific use case.

Quantum sensors for positioning and navigation

Building on the quantum sensing systems outlined above, a range of advanced quantum sensors has been developed and increasingly designed to meet specific application needs. In this section, we review several key quantum sensors that have demonstrated particular promise for enhancing positioning and navigation capabilities due to their distinctive operating principles and inherent advantages. Although the standalone use of quantum sensors in PNT services is not yet a common and mature practice, recent technological advancements have significantly broadened their application potential and enabled a series of innovative demonstrations, which are also examined. Additionally, key specifications are discussed to inform future sensor development and system optimisation. Please note that as quantum sensors build directly on the quantum sensing systems introduced, some conceptual overlap is indeed inevitable, however, this section shifts focus from general principles to specific sensors related to positioning and navigation contexts.

Quantum clocks

Atomic clocks, based on well-characterised and robust transitions in atoms (especially rubidium) represent one of the most mature and widespread quantum systems in use today.

Table 1 Advantages and disadvantages of relevant advanced quantum sensing systems for positioning and navigation including some indicative sensitivities and modalities. Experimentally reported sensitivities show the current state-of-the-art, and future developments should improve these

Quantum Technique	Advantages	Disadvantages	Sensitivity & Modality
Cold Atom Interferometry	Ultra-precise motion measurements No reliance on external signals like GPS Long-term stability	Sensitive to environmental conditions (e.g., temperature) Bulky and expensive Requires precise laser manipulation	Accelerometry 3 μm $s^{-2}/\sqrt{Hz}$ (Salducci et al. 2024) *
			Gyroscope 11 μrad·$s^{-1}/\sqrt{Hz}$ (Salducci et al. 2024) *
			Gravimetry 37 μGal/√Hz (Wu et al. 2019) **
Nitrogen Vacancy (NV) Centres	High sensitivity to magnetic fields Compact and versatile Robust in extreme conditions	Sensitive to temperature and calibration noise	Magnetometry 460 fT/√Hz (Barry et al. 2024)
Optically Pumped Magnetometers	High precision in magnetic field measurements Compact and lightweight Commercially available	Limited by magnetic field strength Environmental sensitivity to temperature changes	Scalar magnetometry 20 fT/√Hz (Oelsner et al. 2019)
Superconducting Quantum Interference Devices (SQUIDs)	High sensitivity in magnetic field and magnetic gradient measurements Potential for high-resolution geomagnetic mapping and magnetic anomaly-based navigation	Requires extremely low temperatures (cryogenics) Large and complex setup Difficult to deploy on mobile platforms	SQUID magnetometry 30–80 fT/√Hz (CSIRO 2024)
			SQUID gradiometer 50–100 pT $m^{-1}/\sqrt{Hz}$ (CSIRO 2024)
Entanglement-Based Sensing	Enhanced precision in phase/field estimation More resilient to certain types of noise	Technically difficult to create and maintain entangled particles Not yet deployed in practical navigation	Magnetometry 12 pT/√Hz (Ruster et al. 2017)
Trapped Ions	Highly accurate time measurements Long-term stability Well utilised for standards and precision metrology	Requires sophisticated and large equipment Sensitive to electric fields and temperature changes	Time, relative uncertainty 9.4×10^{-19} (Brewer et al. 2019)

* A compact, magnetically-launched 87Rb Mach-Zehnder atom interferometer operated at pulse spacing T = 40 ms;

** A mobile caesium Mach-Zehnder atom gravimeter operated at pulse spacing T = 130 ms

They form the backbone of global timekeeping and synchronisation, serving as the temporal foundation of modern PNT architectures through their integration into GNSS (Luccio 2024) and national standards offices (BIPM 2020). Additionally, next-generation clocks provide new opportunities for geodesy and ultra-precise (centimetre-level) height determination through chronometric leveling (McGrew et al. 2018) and may also be implemented in future GNSS (Origlia et al. 2018; Schuldt et al. 2018; Boldbaatar et al. 2023) and terrestrial timing networks (Koelemeij et al. 2022). However, as this paper focuses specifically on positioning and navigation, timing technologies are not discussed in detail. Readers interested in overviews of current and future quantum clocks for PNT are referred to (Ludlow et al. 2015; Jaduszliwer and Camparo 2021; Batori et al. 2021). The rest of this section explores quantum sensors that have been or could be utilised to directly contribute to positioning and navigation capabilities.

Quantum IMUs

Inertial navigation systems are inherently limited by the accumulation of sensor bias and drift errors over time, as indicated above. To address this limitation, quantum inertial navigation systems (QINS), which leverage quantum inertial measurement units (IMUs) including accelerometers and gyroscopes, offer a promising solution for autonomous navigation. Although the overall system architecture of QINS mirrors that of classical INS (Feng 2019), the critical distinction lies in the replacement of conventional inertial sensors with their quantum counterparts. These sensors exploit quantised phase shifts induced by inertial forces like acceleration and rotation, enabling ultra-stable, drift-free measurements with enhanced long-term accuracy.

Among these, quantum accelerometers have seen notable advancements. Cold atom technology forms the core of many such devices, using the wavelike nature of atoms at low temperatures for interferometry. Studies have shown that CAI accelerometers can achieve up to fifty times greater accuracy and greatly reduced drift errors compared

to classical counterparts (Jekeli 2005). Yu et al. (2008) demonstrated 3×10^{-9} g/√Hz at a 100 ms measurement period. Dickerson et al. (2013) also achieved multi-axis inertial sensing using a long time, point-source atom interferometer with spatially resolved detection, reaching a per-shot sensitivity of 6.7×10^{-12} g. This represents an improvement of approximately two orders of magnitude over previous limits. However, this long-baseline configuration is a laboratory demonstration and is not representative of a deployable, mobile quantum accelerometer. In addition, there is growing interest in CAI accelerometers for LEO gravity missions, where their well-defined scale factor and long-term stability can improve the recovery of low-frequency gravity signals. Knabe et al. (2022) assess the prospective benefits of CAI accelerometers for future satellite gravity missions, indicating their potential to complement electrostatic accelerometers, particularly at long integration times. HosseiniArani et al. (2025) further present a hybrid system level concept that combines a CAI with a classical accelerometer to exploit complementary noise characteristics across the measurement band. While these studies focus on gravity field mapping instead of onboard navigation, they remain relevant to resilient PNT, as enhanced gravity models support gravity-aided navigation and the underlying CAI technologies translate to navigation-grade accelerometry in the long term. It is also essential to emphasise that, although quantum accelerometers have greater sensitivity than their classical counterparts, local accelerometers cannot operate without high quality gravity maps. This is due to the equivalence principle, which states that the effects of acceleration and gravity are indistinguishable for local observers. Hence, unravelling kinematic acceleration requires subtraction of the local gravity field from the acceleration measured by the accelerometer. Because the local gravity varies, this subtraction means that at the highest levels of precision, three-dimensional map matching will be required, which is a requirement that applies irrespective of whether the sensor is quantum or classical. In practical cases, accurately determining the local gravitational acceleration is likely to limit overall system performance.

Quantum gyroscopes exploit the phenomena of entanglement and superposition to achieve exceptional sensitivity to angular rotations (Stockton et al. 2011). Notable implementations include atomic Sagnac interferometry and spin precession systems, like nuclear magnetic resonance gyroscopes (NMRGs) and SERF gyroscopes. Recent advancements have led to the development of various quantum gyroscopes (Ledbetter et al. 2012; Jarmola et al. 2021). For example, CAI gyroscopes have demonstrated sensitivities on the order of 10^{-9}–10^{-10} rad $s^{-1}/\sqrt{Hz}$ (Gustavson et al. 2000b; Jentsch et al. 2004). Meanwhile, solid-state approaches, such as the NV-centre-based gyroscope developed by Ledbetter et al. (2012), have achieved sensitivities of $\sim 5 \times 10^{-3}$ rad $s^{-1}/\sqrt{Hz}$ in a 1 mm^3 volume, with the potential to reach 10^{-5} rad $s^{-1}/\sqrt{Hz}$ using dynamical decoupling that mitigates dipolar dephasing. A more recent implementation by Jarmola et al. (2021) demonstrated a diamond nuclear spin gyroscope based on nitrogen-14 spins, achieving a sensitivity of 4.7°/√s ($\sim 8.2 \times 10^{-2}$ rad $s^{-1}/\sqrt{Hz}$) and a bias stability of 0.4°/s ($\sim 7 \times 10^{-3}$ rad/s) under ambient conditions, where the slightly reduced sensitivity reflects both the use of nuclear (rather than electron) spins and the implementation on a rotating platform under practical test conditions. Collectively, these improvements represent a major leap in inertial navigation. By delivering enhanced measurements of acceleration and angular rates, QINS enable more accurate estimation of position, velocity, and attitude, with the long-term stability and precision necessary for reliable operation in GNSS-denied environments.

A different approach to quantum gyroscopes is observed using the London moment. In this case, a spinning superconductor creates a magnetic field aligned with its rotation axis, which can be read out with a magnetometer (London 1951). Such devices have provided some of the most accurate measurements of rotation to date and were employed in the Gravity Probe B mission (Everitt et al. 2011). These systems achieved drift rates on the order of 10^{-11} degrees per hour. However, this required high-precision equipment operated at a temperature of 2 K, rendering the approach impractical for most applications.

Despite the considerable promise of quantum gyroscopes, it is important to stress that such systems currently tend to underperform compared to more conventional optical gyroscopes, especially in real-world scenarios. Optical gyroscopes operate based on the Sagnac effect, which involves measuring the interference between optical fields counterpropagating around a loop (Vali and Shorthill 1976). A key advantage of this approach is that because the two fields are derived from the same source, any intensity noise is removed as a common mode signal. In the absence of rotation, the pathlength for each beam is identical (as they are counterpropagating around the same loop). However any angular rotation breaks the symmetry, which is resolved as a shift in the interference fringes. This output signal can be determined with exceptional precision, enabling highly accurate rotation measurements. For example, commercially available fibre-optic gyroscopes, e.g. the BOREAS D90 (Advanced Navigation 2025) can achieve 0.001 °/√h (2×10^{-5} °/√s) angular random walk error (Song et al. 2023). Similarly, ring laser gyroscopes, which employ an active source as the basis for the interference achieves comparable or superior precision and remain a popular gyroscope technology for applications such as aircraft navigation (Sinelnikov et al. 2024). So, although CAI and London moment

gyroscopes outperform optical systems in terms of sensitivity, significant development is still required before they can replace optical gyroscopes in most practical settings. Indeed, gyroscope development remains an important frontier for quantum sensing.

Superconducting gyroscope solutions are unlikely to challenge classical gyroscopes due to the restrictions of cryogenic fluids and related SWaP-C impact. By contrast, continued advances in CAI could well lead to more practical quantum gyroscopes. A further potentially disruptive technology is quantum optical gyroscopes (Haus et al. 1991; Mehmet et al. 2010; Grace et al. 2020). The advantage of quantum squeezing for interferometry has been identified since at least 1981 (Caves 1981) and has been famously demonstrated in the LIGO experiment (Barsotti et al. 2011). A squeezed light gyroscope operates under almost identical conditions as conventional fibre optical gyroscopes, albeit with the addition of a squeezed light source. Such a source provides light with one quadrature exhibiting noise lower than the standard quantum limit, and by judicious experimental design, the resultant interferometer can be more sensitive than would be expected from standard quantum limits. For these devices to be commercially practical requires further development of cheap and robust sources of squeezed light (Yang et al. 2022).

The performance of quantum inertial sensors is typically characterised by metrics including bandwidth, dynamic range, bias drift, scale factor stability, angular random walk, and sensitivity. For navigation-specific applications, additional specifications including heading accuracy, speed accuracy, and GNSS holdover capability are also critical. Recent studies demonstrate the transformative potential of QINS in enhancing positioning and navigation. For example, Wright (2022) provides a systems-level review of cold-atom inertial sensors, with a particular focus on their practical performance in real-world navigation applications. Battelier et al. (2016) reported a compact cold-atom accelerometer capable of measuring acceleration along two orthogonal axes onboard an aircraft, achieving a one-shot sensitivity of 2.3×10^{-4} g. For illustrative purposes, if this value is treated as a constant single-axis acceleration error persisting over a 60 s period, it would imply an indicative velocity error of ~0.135 m/s and a position drift of ~4.05 m. Note that these back-of-the-envelope values apply only to a single-axis scenario and the actual drift characteristics of a complete quantum IMU are governed by multi-axis coupling and attitude error dynamics and thus differ considerably from this single-axis estimate (Cassel et al. 2023). Subsequently, Battelier et al. (2020) demonstrated a three-axis accelerometer based on matter-wave interferometry for mobile sensing applications, reporting per-shot sensitivities of 10.8 µg (X), 5.5 µg (Y), and 4.4 µg (Z). Under the same assumption of a constant 60 s bias, these values translate to indicative position drifts of ~0.19 m (X), ~0.10 m (Y), and ~0.08 m (Z), respectively.

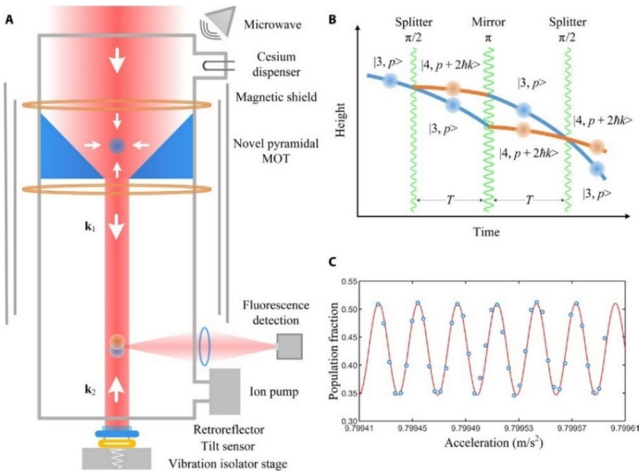


Fig. 3 Mobile atomic gravimeter based on atom interferometry. **a** Schematic of the experimental setup: caesium atoms are loaded into a pyramidal MOT and fall into a fluorescence detection region within a magnetically shielded vacuum chamber. k_1 and k_2 are the wave vectors of the interferometer beams. **b** Mach-Zehnder interferometer sequence using three laser pulses (wavy green lines) to manipulate atomic wave packets (blue and orange lines). **c** Interference fringes at 120 ms with contrast C of 16%; blue dots are single-shot experimental data, and the red curve is a sinusoidal fit. Reproduced with permission from Wu et al. (2019), Science Advances, 5(8): eaax0800, under the CC BY-NC 4.0 license

Quantum gravimeters

Quantum gravity sensors, including gravimeters and gradiometers, are essential tools for detecting subtle spatiotemporal variations in gravitational field. Given that natural gravitational variations typically occur at the level of ~1 µGal (10 nm/s²), achieving such precision requires advanced gravity sensors.

Quantum gravimeters based on CAI address this challenge by measuring phase shifts induced by gravitational acceleration in freely falling atomic clouds, enabling absolute gravity measurements. Under controlled laboratory conditions, short-term sensitivities are typically at the tens-of-µGal/√Hz level, with long-term averaging reaching sub-µGal resolution (Peters et al. 2001; Le Gouët et al. 2008; Merlet et al. 2010; Bodart et al. 2010). In recent years, field-deployable and dynamic-platform demonstrations have also emerged (Geiger et al. 2011; Bidel et al. 2018; Janvier et al. 2022). For example, Ménoret et al. (2018) developed a compact, transportable quantum gravimeter featuring a hollow pyramid reflector and telecom-based laser system, achieving long-term stability better than 1 µGal and a sensitivity of 50 µGal/√Hz. As illustrated in Fig. 3, Wu et al. (2019) showcased a mobile cold atom gravimeter operating

effectively in both laboratory and field settings. The instrument achieved a sensitivity of 37 μGal/√Hz and stability better than 2 μGal over 30 min. Field measurements were completed within minutes, with field uncertainties below 40 μGal.

Apart from gravimeters, recent years have seen notable advancements in quantum gradiometers, which directly measure gravity gradients using two spatially separated interferometers referenced to a common laser source (Mcguirk et al. 2002; Biedermann et al. 2015). This configuration enhances drift suppression and long-term stability, enabling sensitivities several orders of magnitude exceeding those of traditional gradiometers (Douch et al. 2018). Ground-based CAI gradiometers have demonstrated sensitivities of a few tens of 10^{-9} $s^{-2}/\sqrt{Hz}$ (Sorrentino et al. 2014; Asenbaum et al. 2017). Sorrentino et al. (2012) showed that dual-ensemble gradiometer can operate under high-vibration conditions, allowing for accurate gravity gradient measurements even in seismic environments. These instruments are also being explored for space deployment. Compact gradiometer concepts suitable for satellite deployment have been proposed and demonstrated (Carraz et al. 2014; Trimeche et al. 2019), with studies showing their potential for detecting subtle geophysical signals from orbit (Migliaccio et al. 2019). Müller and Wu (2020) demonstrated that a three-arm CAI gradiometer in a GOCE-like orbit could measure all diagonal gravity gradient components with a sensitivity of 3.5×10^{-12} $s^{-2}/\sqrt{Hz}$. Leveraging the extended interrogation time (on the order of tens of seconds) and low-noise environment in microgravity, optimised space-based gradiometers are projected to achieve sensitivities as low as 0.07×10^{-12} $s^{-2}/\sqrt{Hz}$. Such advances are particularly relevant to gravity-aided navigation, where precise gravity gradient data can enhance gravity matching algorithms for position fixing (Jekeli 2006), thereby supporting autonomous and resilient navigation in spaceborne and other remote terrestrial missions. Despite these advancements, transitioning quantum gravity sensors from prototypes to robust, compact, and field-deployable platforms remains a significant challenge, particularly in ensuring high-precision measurements while maintaining sensitivity in a compact, lightweight design (Wu et al. 2014; Ménoret et al. 2018).

Generally, gravitational features are present globally and cannot be jammed or spoofed. This ubiquity supports passive and autonomous position fixing (Jekeli 2006; Wang et al. 2016). For positioning and navigation purposes, evaluating the performance of quantum gravity sensors requires consideration of critical specifications, including sensitivity, bandwidth, dynamic range, stability, matching accuracy, absolute error, update rate, and spatial resolution of the measured gravitational fields. Although classical gravity sensors have long supported navigation and positioning, their standalone use is often constrained by environmental conditions, map resolution, and accuracy limits. In contrast, quantum gravity sensors offer marked improvements in sensitivity and stability and are being tested in various settings. To date, most quantum gravity sensors have operated in static or controlled laboratory environments, with limited demonstrations in dynamic conditions. For example, Bidel et al. (2018) demonstrated sub-mGal absolute gravity measurements (below 10^{-5} m/s^2) aboard a moving ship using CAI gravimeters. Hardman et al. (2016) presented a multifunctional instrument combining a gravimeter, gradiometer, and magnetometer, using a free-falling Bose–Einstein condensate as the test mass. This system achieved a gravitational acceleration precision of $\Delta g/g = 1.45 \times 10^{-9}$. Such sensitivity has the potential to greatly enhance gravity- and gravity gradient-aided navigation by improving the correlation between onboard measurements and prior gravitational maps. As noted by Cassel et al. (2023), while achievable positioning accuracy depends mainly on map availability and resolution, high-quality quantum gravimeters can also contribute to the generation of high-quality gravitational maps. Furthermore, these sensors could support strategic-grade INS by bounding drift and providing accurate gravitational acceleration information. Together, these capabilities underscore the promising role of quantum gravity sensors in enabling resilient positioning and navigation.

Quantum magnetometers

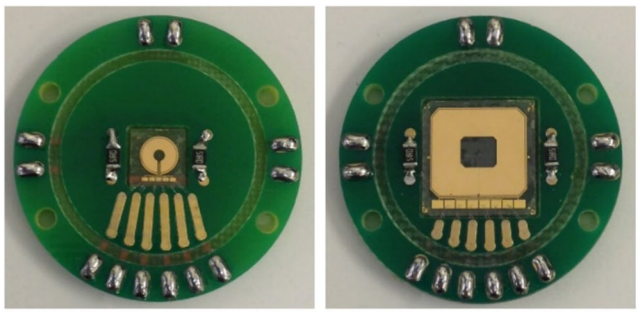
The magnetic field of the Earth also provides a useful geophysical reference for navigation, particularly in GNSS-denied environments. Quantum magnetometers are being developed to leverage this feature for offering external position fixes (Frontera et al. 2018). As described in Table 1, nearly all modern quantum sensing systems have been employed in the development of quantum magnetometers.

SQUID magnetometers continue to offer benchmark performance in sensitivity. For example, Schultze et al. (2008) reported magnetic field resolutions of ~20 fT within a 10 Hz bandwidth, enabling the generation of highly detailed magnetic gradient maps. In magnetic anomaly-based navigation, sensor sensitivity is critical, however, the overall positioning accuracy ultimately depends on the resolution and reliability of the underlying magnetic field maps and platform-induced effects. This indicates the coupled importance of sensor capabilities and environmental priors, similar to the situation in the aforementioned gravity- and gravity gradient-aided navigation. The portability required for most PNT applications mean that low critical temperature magnetometers, such as the aluminium ones used for quantum computing (Nakamura et al. 1999), are not practical due to the requirements for liquid helium cooling or

dilution refrigerators. Nevertheless, high critical temperature SQUIDS, based on Type II superconductors and operating at liquid nitrogen temperatures, are being practically deployed and their usability often more than compensates for their reduced sensitivity. Figure 4 illustrates two standard high-temperature superconducting (HTS) dc SQUID magnetometers, MAG-3 and MAG-8, developed by CSIRO for magnetic sensing applications (Keenan et al. 2013, 2022; CSIRO 2024).

Additionally, optically pumped magnetometers (OPMs) are among the most mature of the emerging quantum sensing technologies, operating at room temperature and demonstrating sensitivities below 20 fT/√Hz under optimal conditions (Oelsner et al. 2019). Their established use in biomagnetic sensing and geophysics has driven wider uptake in applied scenarios. However, OPMs are typically scalar devices, meaning they measure only the magnitude of the field and are prone to "dead zones", orientations in which sensitivity degrades due to alignment between the magnetic field and the optical pumping axis. This limitation poses challenges for orientation-aware navigation. While techniques like balanced optical pumping and vector compensation are being developed to address this issue, scalar OPMs remain best suited for fixed or constrained-orientation applications. Recent work by Muradoglu et al. (2025) showed that, when integrated with real-time denoising and magnetic map-matching algorithms, compact scalar OPMs can support field-deployable magnetic anomaly navigation, achieving up to a ~ 46-fold reduction in positioning error compared to a strategic-grade INS across airborne and ground-based trials.

In contrast, NV-centre diamond magnetometers enable vector magnetic field measurements, allowing simultaneous sensing of all three orthogonal components of the field. This vectorial nature is particularly advantageous in dynamic or orientation-varying navigation scenarios. Operating at room temperature, NV-based systems have seen continuous improvements in sensitivity due to recent innovations.



(a) MAG-3 (b) MAG-8

Fig. 4 Two HTS SQUID magnetometer designs developed by CSIRO. **a** MAG-3, featuring a 3 mm diameter circular pick-up loop; **b** MAG-8, featuring an 8.3 mm × 7.5 mm pick-up loop. Adapted with permission from CSIRO (2024)

These include isotopic enrichment to reduce spin noise (Balasubramanian et al. 2009), integration into optical fibres for increased portability and versatility (Acosta et al. 2010; Bai et al. 2020; Stürner et al. 2021), and the development of laser-threshold magnetometry for high-sensitivity field measurements (Jeske et al. 2016; Dumeige et al. 2019; Hahl et al. 2022). Xie et al. (2020) introduced a dissipative-quantum-sensing protocol that utilises the steady state of NV ensembles for low-frequency magnetic field detection. This approach enhances frequency resolution and sensitivity (1 nT/√Hz above 1 kHz), without requiring state initialisation, complex readout, or dynamical decoupling. While it has not yet reached state-of-the-art sensitivities, this method provides a robust and simplified alternative that outperforms traditional continuous-wave NV methods limited by lock-in amplifier time constants (Schloss et al. 2018; Barry et al. 2016). Most recently, Barry et al. (2024) demonstrated a bulk NV-centre diamond magnetometer with a sensitivity of 460 fT/√Hz in broadband dc-sensitive mode, the highest reported to date without the use of a flux concentrator. This performance was made possible through a combination of low-strain diamond growth, double-quantum magnetometry protocols, and efficient optical readout based on a light-trapping waveguide and balanced photodetection.

Despite these advances, most quantum magnetometers remain at the laboratory stage or in early field-deployable prototypes (Greentree et al. 2025; Graham et al. 2025), and current implementations continue to operate well above their theoretical sensitivity limits. As noted by Bennett et al. (2021), current devices are typically two to three orders of magnitude away from their fundamental sensitivity floors, which might be approached through the exploitation of deeper quantum-mechanical effects. Continued research and development are therefore critical to unlock the full potential of robust quantum magnetometers and to further enhance their performance and reliability in real-world scenarios. Webb et al. (2019) developed a compact, room-temperature NV-centre diamond magnetometer designed for practical low-field sensing in unshielded settings, achieving a sensitivity of ~ 7 nT/√Hz across a 125 Hz bandwidth. While not yet matching laboratory-based state-of-the-art performance, this level of sensitivity is notable for portable applications. When integrated with high-quality magnetic maps and robust matching algorithms, it has the potential to substantially improve positioning accuracy, provided that motion-induced errors and external disturbances are adequately accounted for in dynamic settings. Another example of a field-deployable quantum magnetometer is Lockheed Martin's "*Dark Ice*" prototype (Lockheed Martin 2019), which integrates an NV-centre diamond magnetometer to enable accurate airborne positioning via geomagnetic anomaly measurements. This device features a wide

dynamic range, high bandwidth, and compact size (~30 cm in length and 15 cm in diameter), allowing deployment on small UAV platforms. These prototypes demonstrate the feasibility of field-deployable quantum magnetometers, however, real-world navigation performance is bounded not only by sensor sensitivity but also by the fidelity of magnetic maps and models, as well as by platform corrections.

A key limitation for magnetic navigation lies in map quality and availability. Although extremely high spatial resolution is achievable in principle (Muradoglu et al. 2025), achieving such performance in practice generally requires assumptions about the continuity and interpolation of the available maps. For example, the International Geomagnetic Reference Field (IGRF; Alken et al. 2021) was found to predict the local magnetic field to within 1% (Nair et al. 2025), which is encouraging, but still comparable to many local anomalies and already within reach of existing and near-term quantum sensors. Therefore, high-precision magnetic navigation will require improved magnetic maps and models. This issue is particularly acute in vector navigation, such as that enabled by NV centres. Vector maps of the magnetic anomalies are typically not available, so the vector magnetic field must be derived from scalar maps via the gradient calculations. Such approaches are inherently limited by the resolution of the original map and are further confounded by errors in the attitude of the platform, sensor, or both.

An additional source of complexity is platform corrections. Any metallic object moving in a magnetic field generate its own signature due to eddy currents, in addition to any static magnetic field. This introduces a platform- and velocity-dependent field that is significant relative to the local magnetic field and must therefore be compensated. Such corrections are typically performed via the Tolles-Lawson model (Tolles and Lawson 1950), which has been updated many times (Gnadt et al. 2022). Again, any mismatch between the true magnetic signal and the model compensation will result in positioning error, ultimately limiting the performance available to a quantum magnetometer in the same way as it would a classical magnetometer.

Quantum positioning systems

Unlike individual quantum sensors, quantum positioning systems (QPS), first proposed by Giovannetti et al. (2001), represent a theoretical framework that leverages quantum entanglement and squeezing for positioning, ranging, and synchronisation. Bahder (2004) then refined this concept, and more recent work have explored its potential to outperform classical protocols (Duan et al. 2021; He et al. 2024).

QPS determines space-time coordinates using a method similar to classical time-of-arrival systems but replaces synchronised clocks with frequency-entangled photons and coincidence counting. In theory, this allows enhanced precision and resistance to interference, with conceptual space-based implementations suggesting sub-centimetre accuracy across much of the near-Earth environment. However, QPS remains highly theoretical. Practical implementation would require stable entangled photon pairs, high-efficiency detectors, and robust low-loss quantum channels, through either free-space optics or fibre links. Receiver systems must support ultra-precise time-tagging, and the placement of photon sources, like on satellites or via terrestrial nodes, demands precise alignment and environmental control. In contrast to more mature technologies such as NV-based and optically pumped magnetometers, which are undergoing field trials, QPS has yet to advance beyond laboratory-scale feasibility studies. While promising, its deployment still faces substantial engineering and physical challenges.

Integration of quantum and classical sensors for enhanced PNT

Quantum sensors represent the apex of modern sensing modalities, however, practical PNT architectures require robust and reliable technologies. Consequently, classical systems are expected to remain integral to PNT solutions for the foreseeable future, even if it is "just" as a failsafe. For this reason, hybrid systems that integrate quantum and conventional (classical) sensors have emerged as a promising solution that is likely to be practically important. These integrated architectures are specifically designed to exploit the complementary advantages of both sensor types, compensating for individual limitations and enabling resilient, high-precision PNT capabilities across a variety of applications (Lautier et al. 2014; Cheiney et al. 2018). In terms of specifications, beyond those applicable to quantum sensors, further attention must also be given to compatibility and integration capabilities to fully unlock the benefits of hybrid systems.

In inertial navigation applications, classical sensors, such as MEMS accelerometers and gyroscopes, typically operate at kHz-level sampling rates, allowing them to capture rapid motion dynamics. However, these sensors are affected by long-term biases and scale factor drifts. In contrast, quantum inertial sensors provide unparalleled accuracy, sensitivity, and long-term stability, but are constrained by relatively lower bandwidth, longer measurement cycles, and limited dynamic ranges. These limitations make them prone to missing instantaneous variations during fast-changing conditions (Jekeli 2005). A hybrid quantum-classical INS can effectively mitigate these limitations by leveraging the high bandwidth and dynamic range of classical sensors and the

exceptional stability and accuracy of quantum sensors (Fang and Qin 2012). In such systems, high-rate measurements from classical sensors can be used to anchor or "unwrap" the phase output of quantum sensors, establishing a reliable baseline. Subsequently, data from both sensor types can be integrated based on this baseline, enabling robust and high-precision navigation solutions. Importantly, this integration reduces the need for frequent recalibration using external references such as GNSS or terrestrial beacons. As a result, the hybrid configuration holds significant promise for enabling long-duration, long-distance autonomous navigation missions with enhanced accuracy and resilience (Bongs et al. 2019; Garrido Alzar 2019; Templier 2021).

For example, Cheiney et al. (2018) demonstrated a hybrid navigation system combining a matter-wave interferometer accelerometer with classical accelerometers. This system achieved a precision of 10^{-7} m/s^2 after 11 h of integration, maintaining Schuler oscillations within a 60 mm amplitude, indicating that accelerometer-induced drift can be bounded at the millimetre level over multi-hour operations. Similarly, Wang et al. (2023b) proposed a fusion framework where classical accelerometer readings are used to unwrap quantum phase measurements via a maximum likelihood estimation. In Monte Carlo simulations involving 1000 runs, the system reduced the RMS error of specific force measurements by an order of magnitude compared to a standalone classical accelerometer. Furthermore, it achieved a ~ 50% reduction in RMS velocity error and lowered the position error from several metres (in the classical INS case) to sub-metre levels. Taken together, these findings demonstrate that quantum–classical integration can substantially reduce accelerometer-induced errors, thereby improving positioning and navigation accuracy and reliability in real-world settings. However, it is also worth noting that, in inertial navigation, the mid- to long-term error budget is typically dominated by gyroscope errors and platform alignment, hence, further developments in high precision gyroscopes and judicious use of additional aiding will be essential to enhance overall performance.

Quantum clocks aboard GNSS satellites are another example of hybrid quantum-classical integration. Recent progress in optical clocks technology, characterised by exceptional stability (Schioppo et al. 2017) and ultra-low measurement uncertainty (Nicholson et al. 2015), highlight their transformative potential for next-generation GNSS technology. These clocks have demonstrated clock-related range error of less than 1 mm over a 30-s averaging period, a substantial improvement compared to the ~ 14 mm error associated with the most stable atomic clocks currently deployed on Galileo FOC satellites. Over a 900-second averaging interval, optical clocks have achieved fractional frequency stabilities as low as 1.4×10^{-17}, outperforming existing satellite-based clocks by several orders of magnitude and reducing range errors to the sub-mm levels (Boldbaatar et al. 2023).

Other successful hybrid systems, increasingly adopted for autonomous navigation in GNSS-denied environments, involve incorporating quantum gravimeters and magnetometers into navigation systems (Groves 2015; Canciani and Raquet 2017; Li et al. 2022). Phillips et al. (2022) showed that integrating partial gravity gradient data from quantum gravimeters into a navigation system reduced flight path errors from 4 km (with classical INS-only) to 300–400 m. Wang et al. (2023a) developed a probabilistic map-matching algorithm to support inertial navigation by integrating geophysical measurements with geo-referenced maps. The method addressed map ambiguity issues through probabilistic multiple-hypotheses tracking and achieved a 6% reduction in position error accumulation over a 3.5-hour period compared to standalone precision-grade classical sensors. In a complementary study, Wang et al. (2023c) focused on the use of quantum diamond magnetometers in magnetometry-aided INS. Through simulation tests based on total magnetic intensity (TMI) maps, they assessed navigation performance for a vehicle travelling at a constant 22 m/s at 100 m altitude over 3.6 h. Depending on sensor noise levels, the position errors ranging from 250 to 500 m, highlighting that quantum magnetometers can improve navigation robustness when paired with classical systems, although performance remains sensitive to environmental factors and sensor sensitivity.

Commercial systems, such as Vector Atomic's 8-litre atomic inertial sensor operating in a strapdown configuration (Vector Atomic 2023), have demonstrated micro-g level compatibility with satellite gravity maps during sea trials. Over a 1-hour duration, this level of precision corresponds to a position error of ~ 63 m in the absence of external corrections. The agreement between onboard measurements and satellite gravity models also enables dynamic error correction, improving position accuracy to the sub-metre range during extended missions. Additionally, its onboard optical atomic clocks achieved sub-10 ps precision, maintained over durations 2000 times longer than that of commercial counterparts. In parallel, researchers in Australia have developed the country's first portable NV-diamond vector magnetometer, as shown in Fig. 5 (Kealy and Greentree 2025). Operating at room temperature and in unshielded environments, NV-diamond magnetometer demonstrated a sensitivity of 0.3 nT/$\sqrt{\text{Hz}}$ in recent vehicle-mounted trials (Graham et al. 2025). Airborne testing has also revealed practical challenges and opportunities for magnetometry-aided navigation (Greentree et al. 2024, 2025). Collectively, these advancements illustrate the transformative potential of hybrid quantum–classical systems in enabling precise,



Fig. 5 Portable NV-diamond magnetometer developed by Phasor Quantum. Reproduced with permission from Kealy and Greentree (2025), Navigation News, Royal Institute of Navigation

stable, and resilient navigation across a range of operational scenarios.

Conclusions and discussion

Positioning, navigation, and timing services are indispensable utilities in modern society, underpinning critical infrastructure and operational capabilities across various sectors. While GNSS have traditionally served as the primary solution for absolute positioning and navigation, their inherent vulnerabilities have prompted a paradigm shift towards quantum-enabled alternatives. Quantum sensing systems are emerging as a transformative force for PNT, offering unprecedented precision and resilience, particularly in GNSS-denied or GNSS-contested environments. Recent advances in quantum sensing systems, including cold atom interferometry, atomic vapour cells, SQUIDs, and solid-state quantum systems, demonstrate strong potential to overcome limitations of classical systems. Quantum sensors offer groundbreaking benefits, including unparalleled sensitivity and novel sensing modalities. Practical implementations of these sensors, whether in standalone form or as part of integrated quantum–classical systems, have shown great promise to enhance the accuracy, robustness, and stability of positioning and navigation. Furthermore, we highlight key specifications of quantum sensors relevant to positioning and navigation applications, such as bandwidth, dynamic range and sensitivity, as well as design considerations including calibration traceability, alignment and data-rate compatibility for multi-sensor fusion, and reliance on the availability and quality of reference maps. These factors collectively inform the development and future optimisation of field-deployable systems for real-world positioning and navigation application.

Despite some progress being made in prototyping and early demonstrations, most quantum sensors remain in the early stages of technological and operational maturity. Rather than replacing classical PNT systems, quantum sensors are best positioned as complementary technologies, particularly valuable for enhancing performance and resilience in scenarios where GNSS are unreliable or completely unavailable. At present, the development and broader adoption of these technologies are still constrained by several practical challenges, including:

(1) *Technological limitations*. Decoherence remains a basic challenge to maintaining the stability of entangled quantum states, which are often critical to achieving maximum measurement precision. Moreover, many quantum sensors face limitations in bandwidth, dynamic range, dead-time (e.g., in atom interferometers), and SWaP-C, restricting their utility in dynamic, space-constrained and other challenging operational scenarios. Specifically, while some quantum sensing technologies, such as atomic vapour cells and certain solid-state techniques, have already achieved compact, low-power designs, others, including cold-atom interferometers, SQUIDs, and many optical clock implementations, still require sizeable laser and vacuum subsystems and higher power consumption. Continued miniaturisation and ruggedisation are therefore essential for advancing their practical deployment. In addition, the unique operating principles of quantum sensors also present challenges for establishing robust verification and calibration protocols, crucial for ensuring consistent and reliable performance.
(2) *Environmental and operational constraints*. Many quantum systems require strict environmental conditions, like cryogenic temperatures for superconductors or vacuum and precise laser trapping systems for cold atoms, which limit their functionality on small or mobile platforms. Additionally, environmental noise, mechanical vibrations, and hardware imperfections may further degrade performance, necessitating advanced system designs and robust error correction strategies.
(3) *Integration issues*. Disparities in operational principles, data rates, and modelling assumptions complicate efforts to the seamless integration of quantum and classical systems. In many cases, existing models for platform motion or onboard noise still lack the precision necessary to support quantum systems, requiring the development of new fusion algorithms and control architectures. While most quantum sensors have been validated in laboratory settings, transitioning them to real-world applications will require further innovations in miniaturisation, robustness, and scalability.

Addressing these limitations will require sustained and coordinated efforts across physics, engineering, and systems

integration. Key priorities include enhancing sensor robustness under operational conditions, advancing miniaturised and field-deployable designs, and developing adaptive integration strategies that enable seamless incorporation into existing PNT frameworks. Equally important is the establishment of enhanced protocols and metrics for quantifying the degree of quantum advantage achievable in practical applications. Broadening the deployment of quantum sensors across diverse domains, from airborne and maritime navigation to autonomous vehicles and space systems, will be essential to unlocking their full potential.

In conclusion, while quantum sensors are still in the developmental phase, their unique properties position them as a cornerstone of next-generation PNT systems. Notably, the United Nations proclaimed 2025 as the International Year of Quantum Science and Technology (IYQ), commemorating 100 years since the inception of quantum mechanics (UN General Assembly 2024). Further, the establishment of the joint working group titled "Evaluating the Potential of Next Generation Quantum Sensors for Positioning, Navigation, and Timing" by the International Association of Geodesy (IAG), the Quantum Geodesy initiative, and the International Federation of Surveyors (FIG), marks a significant institutional commitment to advancing this promising research field. Together, these developments offer a timely opportunity for the global research community to accelerate innovation, foster collaboration and engagement, and redefine the future of resilient PNT systems using quantum sensing technologies. This transformative journey ahead holds the promise to revolutionise PNT, paving the way for unprecedented capabilities across scientific, industrial, and strategic domains.

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Declarations

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